

Lesson 26 — Collect the Data (Build the Model)

Matches LEGO Teacher Portal lesson: "Sorting Apples".

Objective (I can...): I can collect simple data with my machine and describe what it shows. **CS**

concept: Collecting, describing, and representing data · **Builds on:** Lesson 25 (we counted repeats; now we gather and describe data). *Honest framing: today kids collect and sort data themselves and watch a computer recognize — no AI is trained by the child.*

AI Connection (2 min, read aloud): AI starts with data — lots of examples collected and sorted. Before a computer can "know" a cat, people collect cat pictures. *Ask the group:* "If we wanted to teach a computer about colors, what data would we collect?"

Materials: LEGO CS & AI kit + device per pair, color cards, a big tally/graph mat per pair. **This is the AI unit's build lesson** — build the AI & Data model now and keep it assembled through Lesson 30.

60-minute flow - Warm-up (10): "Count the room" — pairs collect quick data (how many kids wore red? blue?) with tally marks, then say which group was biggest. That's collecting and describing data. -

Concept + demo (8): The **coach** shows the color sensor reading colors and the device collecting the data (how many reds, how many greens). Describe it together: "We saw more green than red." On-

device, saves nothing. - **Build/Code (25):** Pairs **build the AI & Data model** per the Canvas picture steps. Then they use the color sensor to collect data — show it several color cards and tally how many of each the machine reads — and describe the result on their graph mat (which color came up most?).

Wiggle break + swap at ~12 min. - **Challenge (7):** Build a brick bar-graph (one brick per reading) so the data is easy to see. - **Share + tidy-up (10):** Pairs show their tally and brick graph and describe what the data shows. **Leave the model built.**

Pair roles: Driver builds and runs the sensor, Navigator tallies and describes the data. Swap at the wiggle break. (*Trio rotates build/collect/tally.*)

Differentiation: *Support* — collect just two colors. *Extension / fast-finisher* — collect three colors and say the order from most to least. *Reteach* — read one card at a time and mark the tally together.

Exit check: Pair collects simple data and can describe what it shows (which group had the most). **At-**

home (optional): Collect data at home: count forks vs. spoons in a drawer; tell a grown-up which there are more of. ~5 min.